Core Pure 2 · Chapter 2

Series

The method of differences · Higher derivatives · Maclaurin expansions

Chapter Overview 2

Method of differences

Split a summand into terms which cancel.

Write the first and last terms explicitly.

Retain every uncancelled boundary term.

Maclaurin series

Match derivatives at zero.

Use standard expansions and substitutions.

State the interval of convergence when required.

STARTER – Round 1! 3

1. Find the product

$$\frac{1}{2} \times \frac{2}{3} \times \cdots \times \frac{n-1}{n}$$

1

2. Find

$$\prod_{r=2}^{n} \left(1 - \frac{1}{r^2}\right)$$

2

STARTER – Round 2! 4

Use cancellation to find

$$S_n = \sum_{r=1}^{n} \left(\frac{1}{r} - \frac{1}{r+1}\right)$$

Sum

Example 5

Use the identity
$4r^3 = r^2(r+1)^2 - (r-1)^2 r^2$
to find $\sum_{r=1}^{n} r^3$.

Sum

Partial Fraction Example 6

Find

$$\sum_{r=1}^{n} \frac{1}{4r^2 - 1}$$

Partial fractions

Sum

Test Your Understanding 7

a. Express $2/((2r+1)(2r+3))$ in partial fractions.

b. Hence sum $3/((2r+1)(2r+3))$, for $r = 1, \ldots, n$.

a

b

Harder telescoping sums 8

Find

$$\sum_{r=1}^{n} \left(\frac{1}{r} - \frac{1}{r+2} \right)$$

Sum

1a. Express $2/((r+1)(r+3))$ in partial fractions.

b. Sum from $r = 1$ to n.

c. Evaluate the sum from $r = 10$ to 100.

a

b

c

2. Show that

$$\frac{1}{r(r+1)(r+2)} = \frac{1}{2r(r+1)} - \frac{1}{2(r+1)(r+2)}$$

and hence sum the left side from 1 to n.

Identity

Sum

3. Use partial fractions and the method of differences to sum

$$\sum_{r=1}^{n} \frac{1}{r(r+2)}$$

3

4. Use partial fractions to find

$$\sum_{r=1}^{n} \frac{1}{(r+2)(r+3)}$$

4

5. Prove that $r/(r+1)! = 1/r! - 1/(r+1)!$. Hence find

$$\sum_{r=1}^{n} \frac{r}{(r+1)!}$$

5

6. Show that

$$\frac{2r+1}{r^2(r+1)^2} = \frac{1}{r^2} - \frac{1}{(r+1)^2}$$

and hence sum from 1 to n.

6

8. Find constants a, b such that

$$\sum_{r=1}^{n} \frac{8}{(3r-2)(3r+4)} = \frac{n(an+b)}{(3n+1)(3n+4)}$$

8

9. Find a if

$$\sum_{r=1}^{n}((r+1)^2-(r-1)^2) = an(n+1)$$

9

10a. Write $\sum_{r=1}^{n} 3/((3r+1)(3r+4))$ as $an/(bn+c)$.

b. Sum the same expression from $r = n$ to $2n$.

10a

10b

11. Robin claims

$$\sum_{r=1}^{n} \frac{2r+1}{r(r+1)} = 1 - \frac{1}{n+1}$$

Explain the error in using the method of differences here.

11

12. Find a, b in

$$\sum_{r=1}^{n} \frac{1}{r(r+2)} = \frac{3}{4} - \frac{an+b}{2(n+1)(n+2)}$$

12

13a. Express $4/((2r+1)(2r+5))$ in partial fractions.

b. Evaluate the sum from $r = 16$ to 25, to four decimal places.

a

b

Higher Derivatives 20

Higher derivatives

Differentiate repeatedly. The second derivative is the derivative of the first derivative.

$$f(x), \quad f'(x), \quad f''(x), \quad f^{(n)}(x)$$

$$\frac{dy}{dx}, \quad \frac{d^2y}{dx^2}, \quad \frac{d^ny}{dx^n}$$

Newton notation $\dot{y}, \ddot{y}$ is commonly used for derivatives with respect to time.

Examples 21

1. Find the third derivative of $\ln(1-x)$ at $x = 1/2$.

1

2. For $f = e^{x^2}$, find the first three derivatives at zero.

2

From derivatives to polynomials 22

From derivatives to polynomials

Can a polynomial reproduce the value and derivatives of a function at a chosen point?

At $x = 0$, the coefficients of $a + bx + cx^2 + dx^3 + \cdots$ determine its successive derivatives.

$$f(0) = a, \quad f'(0) = b$$
$$f''(0) = 2c, \quad f'''(0) = 6d$$

Why we might represent functions as power series 23

Why use power series?

Polynomials are easy to evaluate, differentiate and integrate.

A series can approximate functions near a chosen point.

Applications include numerical integration, approximating constants and solving differential equations.

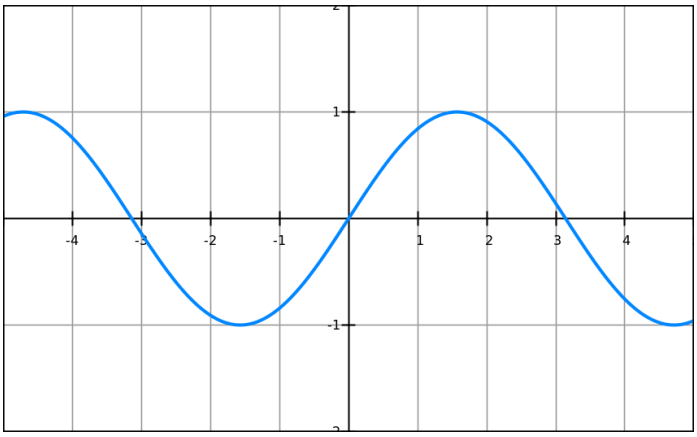
$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$$

At $x = 1$, this gives a series for $\pi/4$.

Now onto the good stuff! 24

Match the value of $\sin x$ at zero with a polynomial.

$f(x) \approx a + bx + cx^2 + dx^3$

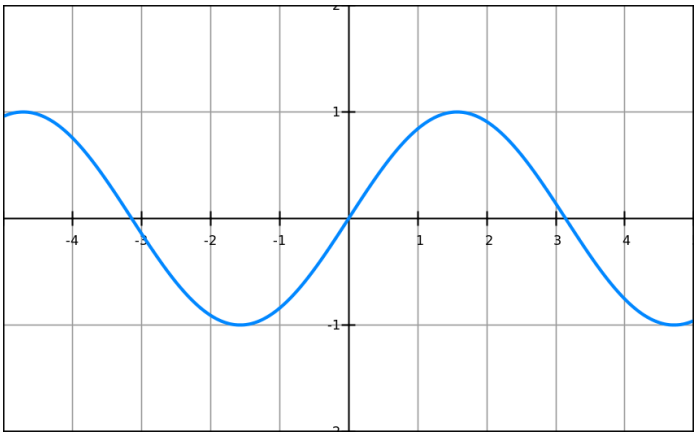


Coefficient

Maclaurin Expansion 25

Now match its first derivative at zero.

$f(x) \approx a + bx + cx^2 + dx^3$

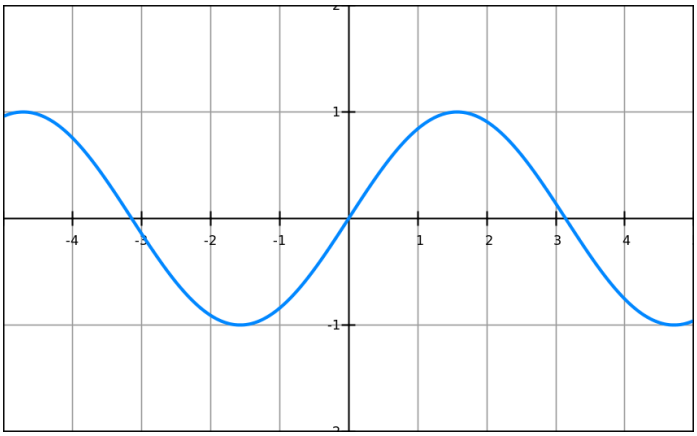


Coefficient

Maclaurin Expansion 26

Now match its second derivative at zero.

$f(x) \approx a + bx + cx^2 + dx^3$

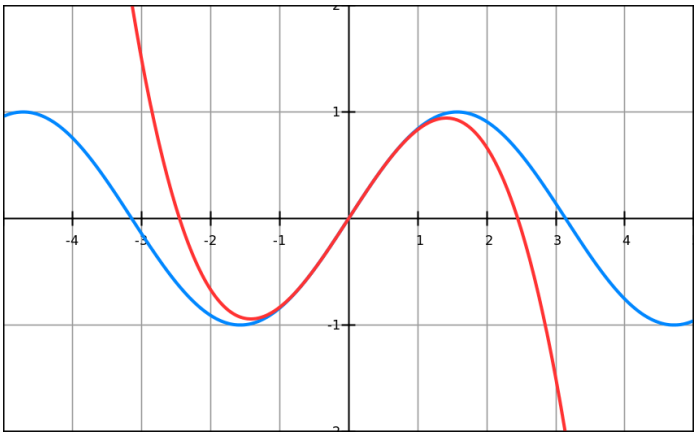


Coefficient

Maclaurin Expansion 27

Match the third derivative at zero.

$f(x) \approx a + bx + cx^2 + dx^3$



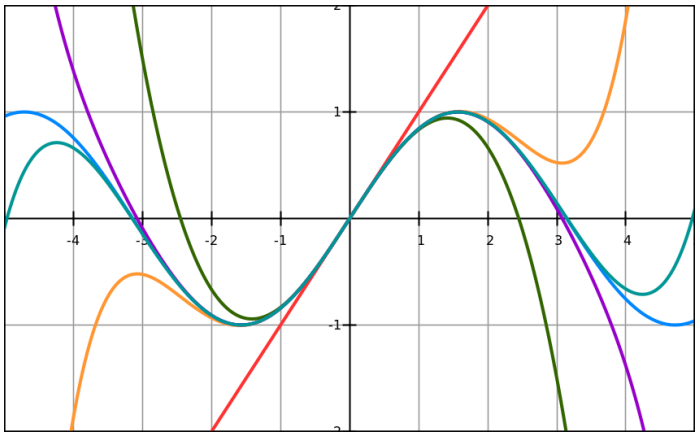
Coefficient

Maclaurin Expansion 28

Maclaurin expansion

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots$$

Equality with an infinite series holds only where that series converges to the function. A truncated series gives an approximation.



$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

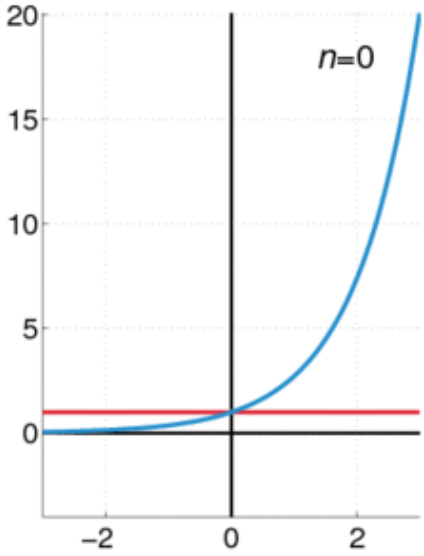
Example 29

Standard series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Both converge for every real x.



The animation shows polynomial approximations improving near the expansion point.

Example 30

Use the first three non-zero terms to estimate $\ln 1.05, \ln 1.25, \ln 1.8$.

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

Estimates

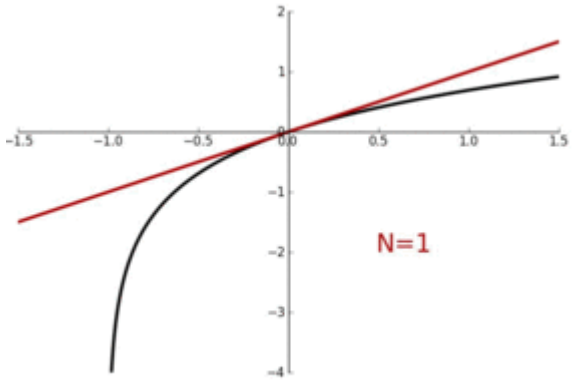
Example 31

Convergence matters

$$\ln(1+x) = \sum_{r=1}^{\infty} \frac{(-1)^{r+1}x^r}{r}$$

$-1 < x \leq 1$

At $x = 1$, the alternating harmonic series converges to $\ln 2$. At $x = -1$, it diverges.



Exercise 2C - Core Pure Year 2 Pages 43-44 32

1a. Obtain the Maclaurin series for $(1-x)^{-1}$.

b. Find the first four terms for $\sqrt{1+x}$.

a

b

2. Find the first three terms of the Maclaurin expansion of $e^{\sin x}$ by differentiation.

2

3a. Derive the Maclaurin series for $\cos x$.

b. Use three non-zero terms to estimate $\cos 30°$.

a

b

6. Find the first five terms in the Maclaurin expansion of $\cos(x - \pi/4)$.

6

7. Let $f(x) = (1 - x)^2 \ln(1 - x)$.
a. Find f' and f''.
b. Evaluate the first three derivatives at zero.
c. Give the first three non-zero Maclaurin terms.

a

b

c

8. Find the first two non-zero terms of $3\sin x - 4x\cos x + x$.

Hence find its ratio to x^3 as $x \to 0$.

a

b

9. Let $f(x) = \ln(\cos x)$.

a. Show $f'(x) = -\tan x$.

b. Find the derivatives at zero up to order four.

c. Give its first two non-zero Maclaurin terms.

a

b

c

10. Find the Maclaurin expansion of $\tan x$, up to and including x^5.

10

Composite Functions 40

Substituting into standard series

$$e^x = \sum_{r=0}^{\infty} \frac{x^r}{r!}$$

$$\sin x = x - \frac{x^3}{3!} + \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \cdots$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$$

Find the first four non-zero terms of $\cos(2x^2)$.

Expansion

Composite Functions 41

Expand to x^3 and state the interval of validity:

$$\ln \frac{\sqrt{1+2x}}{1-3x}$$

Expansion

Validity

Composite Functions 42

Expand $e^{\sin x}$ through the term in x^4, using standard series.

Expansion

2a. Expand $\ln((1+x)/(1-x))$.

b. Hence expand the logarithm of its square root.

c. Estimate $\ln(2/3)$, using three terms.

d. Estimate $\ln 2$, and compare to the calculator value.

a

b

c

d

3. Show that, for small x, $e^{2x} - e^{-x} \approx 3x + 3x^2/2$.

4a. Expand $3x \sin 2x - \cos 3x$ through x^4.

b. Find the limit of $(3x \sin 2x - \cos 3x + 1)/x^2$ as $x \to 0$.

3

4a

4b

6a. Expand $\cos 2x$ through x^8.

b. Hence obtain the expansion of $\sin^2 x$ through x^8.

a

b

7. The first non-zero terms of $\ln(1+x)+(x-1)(e^x-1)$ are px^3+qx^4. Find p, q.

7

8a. Expand $\sin x/(1-x)^2$ through x^4.

b. Find the gradient of this function at the origin.

a

b

10a. Give the first five non-zero terms of $e^{-x^2/2}$.

b. Integrate this approximation to estimate $\int_{-1}^{1} e^{-x^2/2}\,dx$.

a

b

11a. Expand $e^{px} \sin 3x$ through x^3.

b. If $e^{px} \sin 3x + \ln(1 + qx) - x$ has leading term kx^3, find p, q, k.

a

b

12. For $x > 0$, let $f(x) = e^{x - \ln x} \sin x$.

a. Expand, neglecting x^4 and higher powers.

b. Estimate $f(0.1)$ and compare with a calculator.

a

b

13. Let $y = \sin 2x - \cos 2x$.

a. Show $y'' = -4y$, and hence $y^{(4)} = 16y$.

b. Find its first five non-zero Maclaurin terms.

a

b